Proposal: Cloud-Native Multimodal AI Study Assistant (GCP)

1. Problem Statement

This project build a cloud-hosted AI assistant that can answer questions, summarize findings, automate simple tasks, and support voice interaction.

2. Intended Users

People who need support for daily news update, document summary and other simple research task.

3. Cloud Platform and Planned Services

Platform: Google Cloud Platform (GCP)

Planned cloud-native components: - Cloud Run: hosts the assistant backend API. - Cloud Functions: external tool endpoint for automation actions, in our case, a daily news update. - Firestore: lightweight persistent storage for assistant notes/facts. - Cloud Speech-to-Text: transcribes user voice input. - Cloud Text-to-Speech: converts assistant responses to audio. - Artifact Registry + Cloud Build: image storage and CI/CD deployment pipeline.

Cross-cloud/hybrid extension: - Hybrid mode with local client + cloud backend (local user interface calling GCP-hosted API). - LLM provider integration.

4. Type of AI Assistant

This project will implement a multimodal cloud AI assistant with: - Text chat for Q and A, analysis, and summarization. - Voice input/output for accessibility and hands-free use. - Tool-calling support for web search, database lookup and cloud function automation. - Practical use case: summarizing documents, and running simple AI assisted research. - Added automation use case: a daily serverless news briefing workflow that compiles headline summaries into Markdown and delivers them to the webpage.

Proposed Deliverables

- Deployed assistant endpoint running on Cloud Run.
- One AI model/API: Gemini LLM, plus speech services.
- Cloud-native architecture with serverless services and storage.
- Evaluation results using defined test cases.
- Final report with architecture, data flow, metrics, limitations, and future work.

Security and Privacy Approach

Identified risks: - API key leakage from source code or logs. - Command injection through tool-calling shell interface.

Mitigations: - API keys stored in environment variables (or Secret Manager in production), never hard-coded. - Input sanitization and strict allow-list policy for shell commands.

Success Criteria

- Assistant can accept input and return useful AI-generated output.

- Successful daily news update is completed end-to-end.
- Evaluation average score $>=$ 0.70 on rubric-based test set.
- Demonstrated multimodal value over text-only baseline.